Supplementary Material for: Mining Electronic Health Records (EHRs): A Survey

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This online supplement (addendum to the main survey) consists of the research related to various clinical data mining applications as introduced in the main survey (Section 2).

1 CLINICAL DATA MINING APPLICATIONS

1.1 Understanding the Natural History of Disease

Jensen et al. (2014), Teno et al. (2001), Murtagh et al. (2008), Wang et al. (2014), Roque et al. (2011), Doshi-Velez et al. (2014), Wright et al. (2010), Shin et al. (2010), Dunteman (1989), Zhang and Zhang (2002), Cao et al. (2005), Holmes et al. (2011), Zhang et al. (2014), Hanauer et al. (2009, 2013), Gotz (2016), Gotz and Stavropoulos (2014), West et al. (2015), Monsen et al. (2010, 2015), Hanauer and Ramakrishnan (2013), Liao and Chen (2013), Hripcsak (2013), Sacchi et al. (2007), Jin et al. (2008), Batal et al. (2009), Patnaik et al. (2011), Munson et al. (2014), Dasgupta and Chawla (2014), Albers and Hripcsak (2010, 2012), Nachimuthu et al. (2010), Verduijn et al. (2007), Batal et al. (2013), Choi et al. (2016), and Perotte and Hripcsak (2013).

1.2 Cohort Identification

Friedlin et al. (2010), Liao et al. (2010), Halpern et al. (2014), Meystre et al. (2009), Kandula et al. (2011), Rasmussen et al. (2014), Castelli et al. (1977), Newton et al. (2013), Overby et al. (2013), Pathak et al. (2013a), Schram et al. (2014), Boland et al. (2013), Gotz et al. (2014), Wang et al. (2013b), Peissig et al. (2012), Pathak et al. (2012), Ho et al. (2014a, 2014b, 2014c), Schulam et al. (2015), Hu et al. (2012), Lasko et al. (2013), Wang et al. (2015a), Nadkarni et al. (2014), Chen et al. (2015, 2016), Hripcsak et al. (1997), Hripcsak and Albers (2013), Carroll et al. (2011), Xu et al. (2011), Sarvestani et al. (2010), Albers et al. (2014), Liu et al. (2015), Che et al. (2015), and Albers et al. (2014).

1.3 Risk Prediction/Biomarker Discovery

Greenland et al. (2004), Yadav et al. (2016b), Knaus et al. (1991), Yadav et al. (2015a), Sarkar et al. (2012), Letham et al. (2013), Ebadollahi et al. (2010), Feldman and Chawla (2014), Ngufor et al. (2015), Byrd et al. (2014), Kamkar et al. (2015), Tran et al. (2014), Lakshmanan et al. (2013), Vijayakrishnan et al. (2014), Simon et al. (2013), Harpaz et al. (2012), Vellanki et al. (2014), Yadav et al. (2015b), Zhao et al. (2011), Algar et al. (2003), Detsky et al. (1986), Finks et al. (2011), McCrory et al. (1993), Hay et al. (2004), McCarthy et al. (2008), Epstein et al. (1993), Ferguson and Durkin (2002, 2003), Ferguson et al. (1988), SooHoo et al. (2006), Benz et al. (2001), Ozkalkanli et al. (2009), Propst et al. (2000), Klastersky et al. (2006), Liu and Hauskrecht (2013, 2015), Liu et al. (2013a, 2013b), Panahiazar et al. (2014), Gotz et al. (2011), Miotto et al. (2016), Reddy and Aggarwal (2015), Paxton et al. (2013a), Wilcox and Hripcsak (2003), Stiglic et al. (2015), Liang et al. (2014), Lin and Haug (2008), Zolfaghar et al. (2013), Caruana et al. (2015), Westra et al. (2011), Skevofilakas et al. (2010), Oztekin et al. (2009), Ghalwash and Obradovic (2014), Somanchi et al. (2015), Li et al. (2016a, 2016b, 2016c), Cheng et al. (2016), Collins et al. (2011), Simon et al. (2011), Mani et al. (2014), Matheny et al. (2010), Pakhomov

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et al. (2011), Fung et al. (2008), Wells et al. (2008), Vinzamuri et al. (2014), Li et al. (2015), van der Heijden et al. (2014), Jackson et al. (2011), Paxton et al. (2013b), Maroco et al. (2011), Breault et al. (2002), Wang et al. (2015b), Sun et al. (2014), Wang et al. (2014), Yu et al. (2008), Ghassempour et al. (2014), Gatti et al. (2011), Sandri et al. (2014), Vinzamuri and Reddy (2013), Peelen et al. (2010), Choi et al. (2015), Wang et al. (2013a), Luo et al. (2016), Hripcsak et al. (2015), Hagar et al. (2014), and Dey et al. (2015).

1.4 Quantifying the Effect of Intervention

Schrom et al. (2013), Campbell (1996), Westra et al. (2017), Prochaska et al. (2008), Ronsmans and Campbell (2011), Law et al. (2003), Pruinelli et al. (2016), Yadav et al. (2016a), Schrom et al. (2013), Westra et al. (2017), Yadav et al. (2016a), and Yadav (2017).

1.5 Constructing Evidence-Based Guidelines

Liu and Rubin (2012), Eibling et al. (2014), Agrawal and Kosowsky (2009), Zeng et al. (2014), Pruinelli et al. (2014), Heselmans et al. (2013), Bidra (2014), Goh et al. (2014), Pruinelli et al. (2015), Pivovarov et al. (2014, 2014), Mani et al. (2007), and Pruinelli et al. (2016).

1.6 Adverse Event Detection

Vilar et al. (2012), Harpaz et al. (2012), Epstein et al. (2013), Iyer et al. (2014), Haerian et al. (2012), White et al. (2013), Pathak et al. (2013c), Shen et al. (2014), Nachimuthu et al. (2010), Rose et al. (2005), Davis et al. (2014), van der Velde et al. (2012), Khan et al. (2012), Sathyanarayana et al. (2014), Iyer et al. (2014), Pathak et al. (2013b), Carroll et al. (2011), Xu et al. (2011), Bobo et al. (2014), Zhang et al. (2015), Melton and Hripcsak (2005), Hauskrecht et al. (2013), and Westra et al. (2017).

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